

It's a Big World Out There¹

Your name:

Who you are working with:



Warm-Up

1. Suppose we know that

$$10^2 = 10 \cdot 10 = 100$$

$$10^3 = 10 \cdot 10 \cdot 10 = 1000$$

What are the following equal to? Feel free to leave your answer as an exponent (so that it looks like 10^n for some n)

(a) $10^4 =$

(b) $10^2 \cdot 10^4 =$

(c) Let's try something a bit bigger,² try keeping your answer in exponent form: Simplify $10^9 \cdot 10^{12} =$

(d) So if I have two whole numbers a and b what would $10^a \cdot 10^b$ be equal to?

(e) Let's try something interesting: What would $10^{\frac{1}{2}} \cdot 10^{\frac{1}{2}}$ be equal to? Think about the previous questions, and what the rule you've been using (in your head) to solve all these...

2. Now let's have some fun, given that $(10^2)^2 = 10^2 \cdot 10^2 = 10000 = 10^4$

(a) What can we say about $(10^3)^3$?

(b) What about $(10^2)^2 \cdot 10^2$?

(c) Can you simplify $((10^2)^2)^2$ and $((10^3)^3)^3$?

(d) With those questions in mind, let's say we have two numbers a and b just like before, how could we rewrite $(10^a)^b$ as something that looks like 10 to some power?

¹An introduction to Scientific Notation and Scale in five parts, written by Joshua Aaron, Source for Comic, *It's a Magical World* by Bill Watterson.

²The answer to this question (in bytes) is equivalent to the approximate size of the internet! That is, if this article's sources are sensible: <https://en.wikipedia.org/wiki/Zettabyte>

Asking the Big Questions

Okay, now that we have warmed up and are clearer about the rules for exponents, let's ask a couple questions here. **How big is planet earth?** Now, before you answer, let's think about the question for a moment, do we mean how much space it takes up? That would mean we would be looking for the earth's *Volume*. Do we mean how much it weighs? Well, that could be hard, since we don't have a giant scale in space—and gravity might be a problem so let's change that to *how much stuff* is in the Earth, that would be asking about the Earth's *Mass*. And finally, since we live on the Earth³ how much space is there for us to live? This would be asking about the Earth's *surface area*. But you know all this of course right? So what do you think? How big is the Earth? Take a guess! You might be surprised by the answer.

3. What is your guess for the *volume* and *surface area* of the Earth? How⁴ did you come to that number?

4. Suppose we looked went online and found that if we were to model the Earth with a sphere it would have radius $r = 6371km$. With that in mind, what is its volume in m^3 ? How did you calculate it? if you wanted to approximate it without a calculator, how would you do it?

5. With the radius r we can also calculate the *surface area* of our approximately spherical Earth in m^2 : What does that calculation look like? If you wanted to approximate it without a calculator, how would you do it?

6. Now *mass* is another story right? What would you guess? Probably something huge right? How did you reason to get your guess?

7. For the *mass* we need another piece of information: suppose the average density of the Earth is $d = 5.51g$ per cm^3 , with that in mind (and some thoughts about what density means) what would we calculate Earth's mass to be? Any chance of trying that without a calculator?

³If you do not live on the earth, come see me—I am so sorry for the long commute!

⁴*Hint:* it might be helpful to first guess the *radius* of the Earth and go from there...